

G.1. Examples

We illustrate the task setup with a single example per task, showing the prompt formatting and a cherry-picked example. In each case, we show outputs from a dense Llama 2 13B model, SparQ Attention ($r = 8, k = 128$), H₂O and LM-Infinite ($k = 192$). Where “...” appears, we have truncated the line of text for brevity.

G.1.1. QUESTION ANSWERING (SQUAD 1-SHOT)

```
### PROMPT (5725e1c4271a42140099d2d9)
Title: University of Chicago. Background: Current ...
Title: Harvard University. Background: Harvard has...
Title: Oxygen. Background: In one experiment, Lavo...
Title: Oxygen. Background: Oxygen storage methods ...
Title: Fresno, California. Background: This vibran...
Title: Fresno, California. Background: Before Worl...
Title: Steam engine. Background: The working fluid...
Title: Sky (United Kingdom). Background: While BSk...
From what you've just read about Fresno, California, please answer the following questions.
Question: Where is Audra McDonald from?
Answer: Fresno
Question: In what year did Roger Rocka's Dinner Theater & Good Company Players open?
Answer:

### OUTPUT
    DENSE: 1978
    SPARQ: 1978
    H2O: 1979
LM-INFINITE: 1975 (Roger Rock
```

G.1.2. QUESTION ANSWERING (TRIVIAQA 0-SHOT)

```
### PROMPT (dpql_5685)
Apéritifs and digestifs ( and) are drinks, typical...
Apéritifs
An apéritif is an alcoholic beverage usually serve...
"Apéritif" may also refer to a snack that precedes...
"Apéritif" is a French word derived from the Latin...
...
* Distilled liquors (ouzo, tequila, whisky or akva...
* Liquor cocktails (Black Russian, Rusty Nail, etc...
In certain areas, it is not uncommon for a digesti...
Bitter digestifs typically contain carminative her...
In many countries, people drink alcoholic beverage...
Question: Which aperitif is named for the Paris chemist who created it in 1846?
Answer:

### OUTPUT
    DENSE: Dubonnet
    SPARQ: Dubonnet
    H2O: Dubonnet
LM-INFINITE: Byrrh
```

Note that for Pythia, the prompt “Single-word answer:” was used in place of “Answer:”, as this helped prevent the model from restating the question in the answer (often qualitatively correct, but not a regex match).

G.1.3. SUMMARISATION (CNN/DAILYMAIL)

PROMPT (a62bbf503be06e8b1f8baa4f3cd537310d5aa3bc)

Article: Prince William arrived in China tonight for one of the most high-profil...

Summary:

OUTPUT

DENSE: Prince William arrived in China tonight for one of the most high-profile { ...

SPARQ: Prince William arrived in China tonight for one of the most high-profile { ...

H2O: Prince William arrived in China tonight for a high-profile visit that will ...

LM-INFINITE: Prince William and Kate Middleton are in Japan for a three-day tour. The ro...

G.1.4. REPETITION (SHAKESPEARE)

PROMPT (210496)

you mistake me much;

I do lament the sickness of the king.

...

...

Peace, children, peace! the king doth love you well:

Incapable and shallow innocents,

You cannot guess who caused your father's death.

Boy:

Grandam, we can; for my good uncle Gloucester

Told me, the king, provoked by the queen,

Devised impeachments to imprison him :

And when my uncle told me so, he wept,

And hugg'd me in his arm, and kindly kiss'd my cheek;

...

...

the king doth love you well:

Incapable and shallow innocents,

You cannot guess who caused your father's death.

Boy:

Grandam, we

OUTPUT

DENSE: can; for my good uncle Gloucester

SPARQ: can; for my good uncle Gloucester

H2O: can;

LM-INFINITE: 'll not stand to prate, but to the purpose.

G.1.5. LANGUAGE MODELLING (WIKITEXT-103)

QUERY (2)

= Mellor hill fort =

Mellor hill fort is a prehistoric site in North West England , that dates from ...

= = Location = =

Mellor lies on the western edge of the Peak District in the Metropolitan Boroug...

= = Background = =

Until the 19th century little was known about hill forts ; none had been excava...
The study of hill forts was popular in the 19th century , with a revival in the...

= = History = =

There is evidence of human activity on the site pre @-@ dating the Iron Age , a...
A flint dagger was discovered on the site . This type of artefact is rare in Gr...
The hill fort was built in and used throughout the Iron Age , as demonstrated b...

Fragments of glass , possibly Roman in origin , and shards of pottery which date to the 1st and 2nd centuries AD , indicate the site was used in the Romano @-@ British period . However [no Roman structures have been discovered](#) , and the nature of Roman activity at the site is a source of speculation . The position of the hilltop indicate that it was easily defended ; however , local finds indicate it was a high @-@ status settlement rather than a military outpost unless a similar feature was located nearby . One reason that Roman structures have not been identified is that the Romano

BPC

DENSE: 0.669

SPARQ: 0.673

H2O: 0.685

LM-INFINITE: 0.692